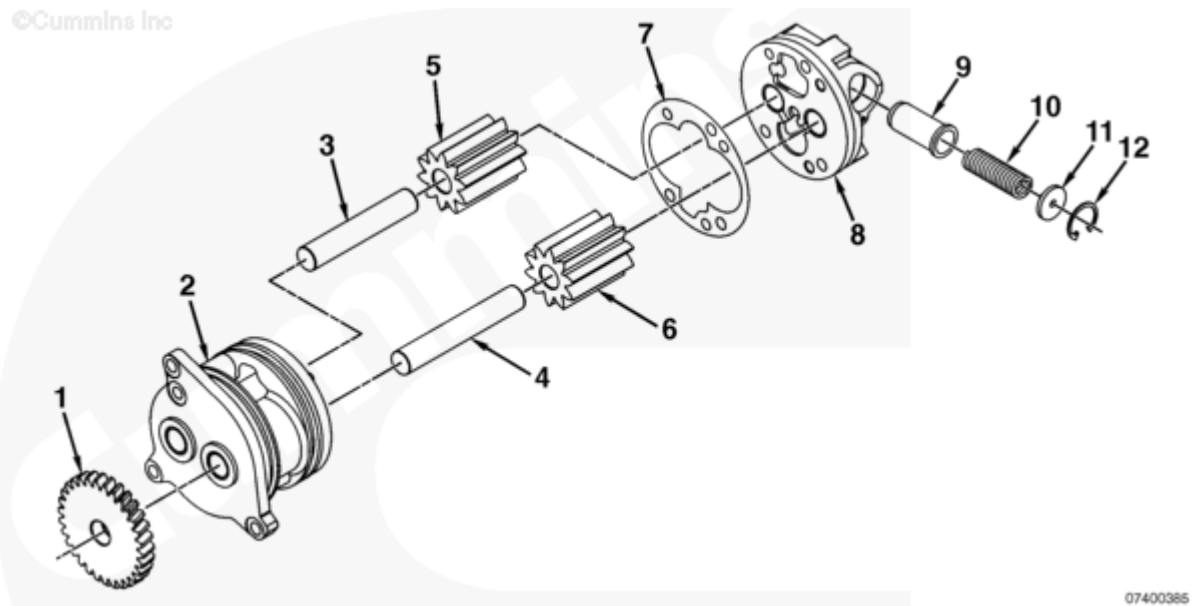


Trainings ▾

Exploded View



1. Lubricating pump gear
2. Lubricating pump body
3. Idler shaft
4. Lubricating pump drive shaft
5. Idler pump gear
6. Lubricating pump gasket
7. Dowel
8. Drive pump gear
9. Lubricating pump cover and bushing
10. Retainer
11. Pressure regulator spring
12. Pressure regulator plunger
13. Snap ring.

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

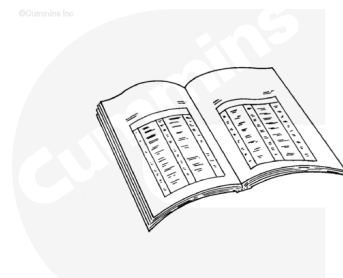
⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when draining oil or replacing filters that the oil is not spilled or drained into the bilge area. The oil and oil filters must be disposed of in accordance with local environmental regulations.

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-025-tr.html>)



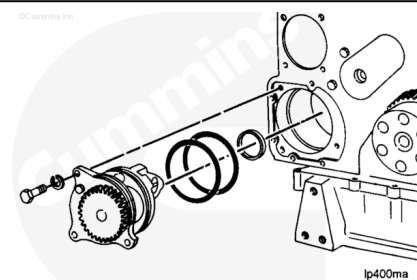
ck800wa

Remove

Remove the three capscrews from the oil pump mounting flange.

Use a pry bar and gently pry the oil pump out of the engine block.

Remove and discard the two o-rings and the seal ring from the lubricating oil pump.

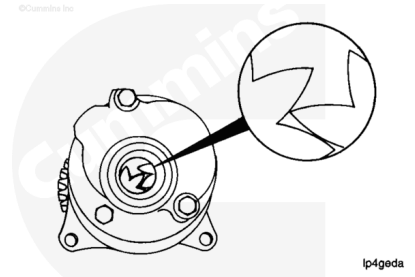


lp400ma

Inspect for Reuse

⚠ WARNING ⚠

When using solvents acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the oil pump body with solvent and dry with compressed air.

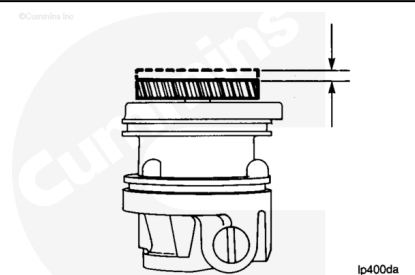
Flush the oil pump with solvent.

Rotate the gears and inspect for damage.

If the gears are damaged, the oil pump **must** be reconditioned or replaced.

Measure the oil pump end clearance.

Oil Pump End Clearance		
mm		in
0.102	MIN	0.004
0.254	MAX	0.010



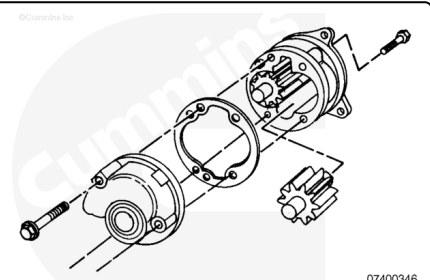
If the oil pump is **not** within specifications, the oil pump **must** be reconditioned or replaced.

Disassemble

Remove the four capscrews.

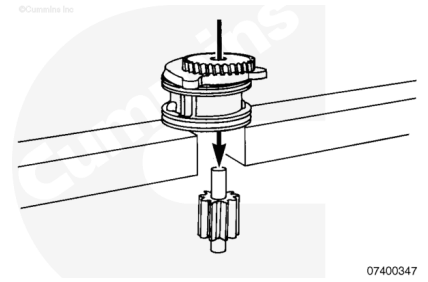
Tap on the dowel with a mallet to separate the cover from the oil pump housing.

Remove oil pump cover, idler gear, and shaft from housing.



Support the housing on an arbor press.

Press the shaft through the drive gear.



07400347

⚠ WARNING ⚠

To reduce the possibility of personal injury, hold the regulator stop in position when removing the roll pin. The regulator stop is under pressure and can pop out.

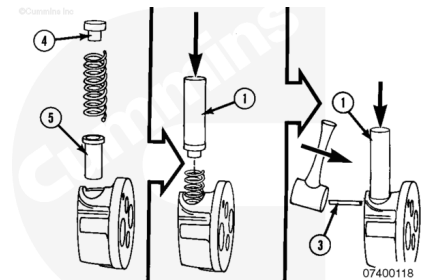
Use an arbor press and mandrel (1) to hold the retainer (4) in place while the dowel pin is being removed.

Some models use a short dowel and roll pin or a snap ring to hold the retainer.

Remove the dowel pin (3) with a brass drift (2).

Slowly raise the arbor press to relieve the spring tension.

Remove the retainer (4), spring, and regulator plunger (5).

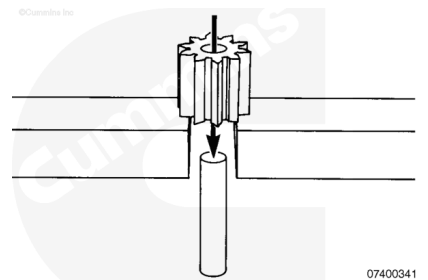


07400118

Note : Only disassemble the gear and shaft when the gear or shaft must be replaced.

Support the gear and shaft in an arbor press.

Press the shaft through the gear.

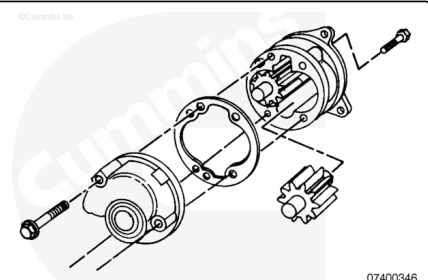


07400341

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



07400346

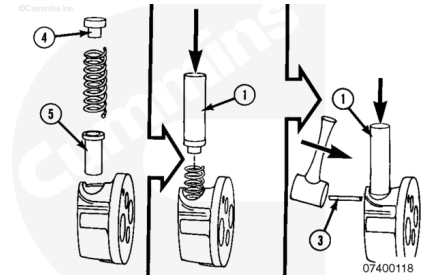
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the oil pump parts with solvent and dry with compressed air.

Inspect the regulator plunger (5), spring, and retainer (4) for damage.

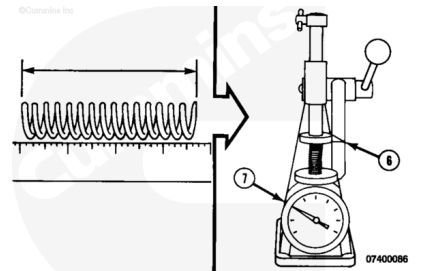
If the regulator plunger, spring or retainer are damaged, the damaged component **must** be replaced.



Check the free length of the spring.

Spring Free Length (Approximately)

mm	in
95	3.75



Measure the spring force (7) at the working height (6) with valve spring tester, Part Number 3375182.

Spring Working Height

mm	in
63.88	2.515

Spring Force

n		lb
320	MIN	72
342	MAX	77

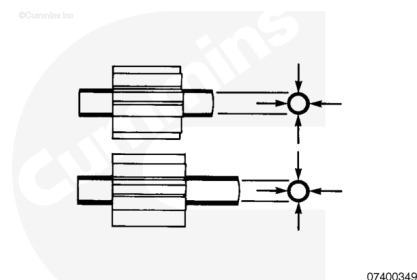
If the spring is **not** within specifications, it **must** be replaced.

Inspect the gears for damage.

If the gears are damaged they **must** be replaced.

Measure the oil pump shaft outside diameter.

Oil Pump Shaft Outside Diameter		
mm		in
22.212	MIN	0.8745
22.225	MAX	0.8750

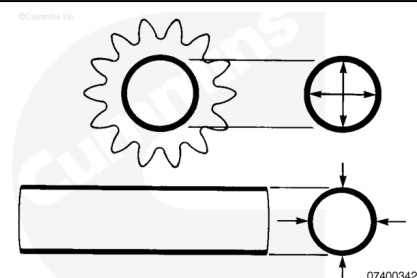


If the oil pump shaft is **not** within specifications, the oil pump shaft **must** be replaced.

Note : Do not disassemble the gear and shaft to make this measurement. Only measure the gear inside diameter if the shaft is replaced.

Measure the oil pump gear inside diameter.

Oil Pump Gear Inside Diameter		
mm		in
22.174	MIN	0.8730
22.187	MAX	0.8735



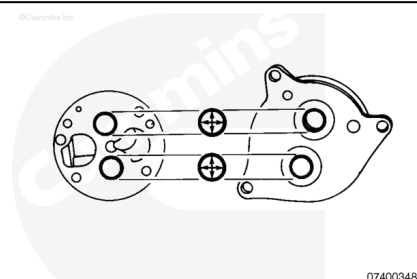
If the oil pump gear inside diameter is **not** within specifications, the oil pump gear **must** be replaced.

Inspect the oil pump housing and cover for damage.

If the oil pump body or cover are damaged, the damaged component **must** be replaced.

Measure the inside diameter of the oil pump bushings.

Oil Pump Bushing Inside Diameter		
mm		in
22.263	MIN	0.877
22.324	MAX	0.879



If the oil pump bushings are **not** within specifications, the bushings **must** be replaced.

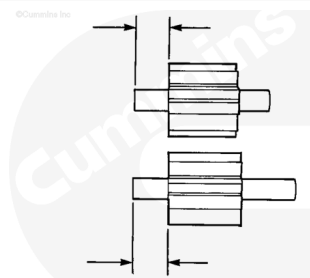
Replace the oil pump bushings from the oil pump body and or oil pump cover.

Assemble

Lubricate the shaft and gear with engine oil.

Install the gear onto the shaft with an arbor press.

Gear Location on Shaft				
	mm		in	
Idler Gear	3.30	MIN	0.130	
Drive Gear		3.81	MAX	0.150



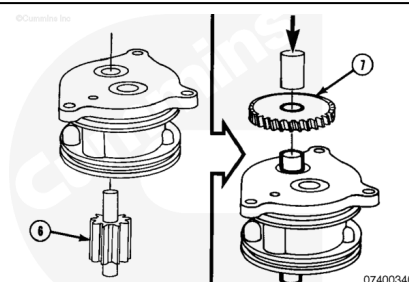
Lubricate the parts with engine oil.

Install the drive gear and shaft assembly (6) into the housing.

Support the shaft in an arbor press.

Place the gear (7) over the shaft with the part number positioned up.

Use a mandrel and the arbor press to install the gear until it is even with shaft.



Lubricate the bushings with engine oil.

Install the shaft into the housing.

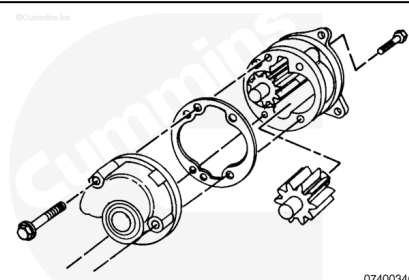
Install the gasket.

Install the cover onto the dowels. Drive the cover onto the dowels with a mallet.

Install the capscrews.

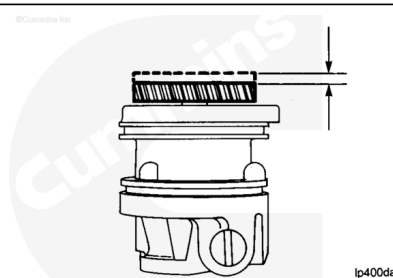
Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



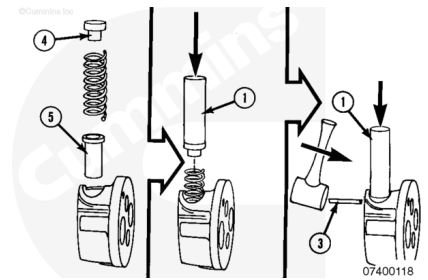
Rotate the assembly and measure the end clearance.

Oil Pump End Clearance			
mm			in
0.102	MIN		0.004
0.254	MAX		0.010



If the oil pump end clearance is **not** within specifications, the oil pump **must** be reconditioned or replaced.

Install the regulator plunger, spring, retainer, and snap ring (or dowel pin).



Install

Apply a light coat of grease to the seal ring to hold it in place during assembly.

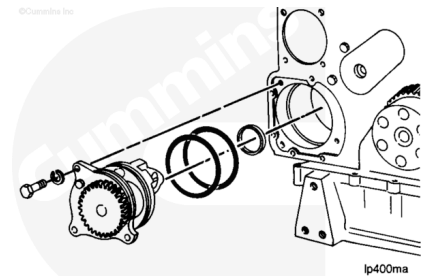
Install the o-rings and the seal ring onto the pump.

Lubricate the o-rings and bore in the block with vegetable oil.

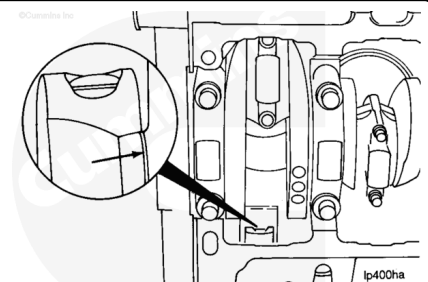
Install the pump and the capscrews.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Look between the number one and number two main bearing caps. The seal **must** be visible in the space between the rear of the pump and the engine block.



Finishing Steps

- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/18/18-007-025-tr.html>)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section .7

(/qs3/pubsys2/xml/en/procedures/18/18-007-037-tr.html)

- Operate the engine and check for leaks.



Last Modified: 13-Feb-2026
